

Newman-Penrose Formalism Calculations

Stephani et al., Equation (21.61)

Metric

$$ds^2 = -\frac{x^2}{N^2}dt \otimes dt - \frac{n}{N^2}dt \otimes d\varphi + \frac{x^2}{(\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2)N^2}dx \otimes dx - \frac{n}{N^2}d\varphi \otimes dt + \left(\frac{\epsilon x^2 + b \log(x) + c}{N^2}\right)d\varphi \otimes d\varphi + \frac{1}{N^2}dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, \varphi, z)$$

Metric Functions

$$N(z) = N$$

$$G(x) = \epsilon x^2 + b \log(x) + c + \frac{n^2}{x^2}$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}x}{2N}\right) dt + \left(-\frac{\sqrt{2}\sqrt{\epsilon x^2 + b \log(x) + c + \frac{n^2}{x^2}}}{2N} + \frac{\sqrt{2}n}{2xN}\right) d\varphi$$

$$n = \left(\frac{\sqrt{2}x}{2N}\right) dt + \left(\frac{\sqrt{2}\sqrt{\epsilon x^2 + b \log(x) + c + \frac{n^2}{x^2}}}{2N} + \frac{\sqrt{2}n}{2xN}\right) d\varphi$$

$$m = \left(\frac{1}{\sqrt{2\epsilon x^2 + 2b \log(x) + 2c + \frac{2n^2}{x^2}}N}\right) dx + \left(\frac{i\sqrt{2}}{2N}\right) dz$$

$$\bar{m} = \left(\frac{1}{\sqrt{2\epsilon x^2 + 2b \log(x) + 2c + \frac{2n^2}{x^2}N}} \right) dx + \left(-\frac{i\sqrt{2}}{2N} \right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}\epsilon x^4 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}bx^2 N \log(x)}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}cx^2 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}n^2 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} \right. \\ \left. + \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2 nN}}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} \right) dt + \left(-\frac{\sqrt{2}xN}{2\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}} \right) d\varphi$$

$$n = \left(-\frac{\sqrt{2}\epsilon x^4 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}bx^2 N \log(x)}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}cx^2 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}n^2 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} \right. \\ \left. - \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2 nN}}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} \right) dt + \left(\frac{\sqrt{2}xN}{2\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}} \right) d\varphi$$

$$m = \left(\frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2 N}}{2x} \right) dx + \left(\frac{1}{2}i\sqrt{2}N \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2 N}}{2x} \right) dx + \left(-\frac{1}{2}i\sqrt{2}N \right) dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}\epsilon x^4 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}bx^2 N \log(x)}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}cx^2 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} \\ - \frac{\sqrt{2}n^2 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} + \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2 nN}}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} \partial_t - \frac{\sqrt{2}xN}{2\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}} \partial_\varphi$$

$$\Delta = -\frac{\sqrt{2}\epsilon x^4 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}bx^2 N \log(x)}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}cx^2 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} \\ - \frac{\sqrt{2}n^2 N}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} - \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2 nN}}{2(\epsilon x^5 + bx^3 \log(x) + cx^3 + n^2 x)} \partial_t + \frac{\sqrt{2}xN}{2\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}} \partial_\varphi$$

$$\delta = \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}N}{2x} \partial_x + \frac{1}{2}i \sqrt{2}N \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}N}{2x} \partial_x - \frac{1}{2}i \sqrt{2}N \partial_z$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\begin{aligned} \kappa = & \frac{\sqrt{2}\epsilon n x^4 N}{2(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} + \frac{\sqrt{2}bnx^2 N \log(x)}{2(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} - \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}bx^2 N \log(x)}{4(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} + \frac{\sqrt{2}cnx^2 N}{2(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} \\ & + \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}bx^2 N}{8(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} - \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}cx^2 N}{4(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} + \frac{\sqrt{2}n^3 N}{2(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} - \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}n^2 N}{2(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} \end{aligned}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\begin{aligned} \nu = & \frac{\sqrt{2}\epsilon n x^4 N}{2(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} + \frac{\sqrt{2}bnx^2 N \log(x)}{2(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} + \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}bx^2 N \log(x)}{4(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} + \frac{\sqrt{2}cnx^2 N}{2(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} \\ & - \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}bx^2 N}{8(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} + \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}cx^2 N}{4(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} + \frac{\sqrt{2}n^3 N}{2(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} + \frac{\sqrt{2}\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}n^2 N}{2(\epsilon x^6 + bx^4 \log(x) + cx^4 + n^2 x^2)} \end{aligned}$$

$$\alpha = -\frac{1}{4}i \sqrt{2} \frac{\partial}{\partial z} N + \frac{\sqrt{2}nN}{4x^2}$$

$$\beta = -\frac{1}{4}i \sqrt{2} \frac{\partial}{\partial z} N + \frac{\sqrt{2}nN}{4x^2}$$

$$\begin{aligned}\tau = & \frac{i\sqrt{2}\epsilon x^4 \frac{\partial}{\partial z} N}{2(\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2)} + \frac{i\sqrt{2}bx^2 \log(x) \frac{\partial}{\partial z} N}{2(\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2)} - \frac{\sqrt{2}\epsilon x^2 N}{2\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}} + \frac{i\sqrt{2}cx^2 \frac{\partial}{\partial z} N}{2(\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2)} \\ & - \frac{\sqrt{2}bN \log(x)}{4\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}} + \frac{i\sqrt{2}n^2 \frac{\partial}{\partial z} N}{2(\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2)} - \frac{\sqrt{2}bN}{8\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}} - \frac{\sqrt{2}cN}{4\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}}\end{aligned}$$

$$\begin{aligned}\pi = & \frac{i\sqrt{2}\epsilon x^4 \frac{\partial}{\partial z} N}{2(\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2)} + \frac{i\sqrt{2}bx^2 \log(x) \frac{\partial}{\partial z} N}{2(\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2)} + \frac{\sqrt{2}\epsilon x^2 N}{2\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}} + \frac{i\sqrt{2}cx^2 \frac{\partial}{\partial z} N}{2(\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2)} \\ & + \frac{\sqrt{2}bN \log(x)}{4\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}} + \frac{i\sqrt{2}n^2 \frac{\partial}{\partial z} N}{2(\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2)} + \frac{\sqrt{2}bN}{8\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}} + \frac{\sqrt{2}cN}{4\sqrt{\epsilon x^4 + bx^2 \log(x) + cx^2 + n^2}}\end{aligned}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = \frac{bN^2}{4x^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = -\frac{1}{2}\epsilon N^2 - \frac{1}{2}N \frac{\partial^2}{(\partial z)^2} N$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{4}\epsilon N^2 + \frac{1}{4}N \frac{\partial^2}{(\partial z)^2} N + \frac{bN^2}{8x^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{1}{2}\epsilon N^2 - \frac{1}{2}N\frac{\partial^2}{(\partial z)^2}N$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{bN^2}{4x^2}$$

$$\Lambda = -\frac{1}{4}\epsilon N^2 - \frac{1}{2}\frac{\partial}{\partial z}N^2 + \frac{1}{4}N\frac{\partial^2}{(\partial z)^2}N - \frac{bN^2}{24x^2}$$

Weyl Tensor Components

$$\Psi_0 = \frac{bN^2}{4x^2}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{bN^2}{12x^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{bN^2}{4x^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "*D*"